MCP-referenced timing calibration of CMS BTL

CMS

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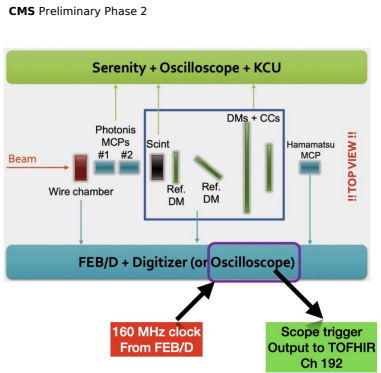
Discussion draft; not approved for public release.

Abstract

This study examines a timing-calibration procedure for CMS barrel timing layer (BTL) prototype scintillator bars using September 2025 beam-test data. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and locally weighted smoothing map the reference into the BTL time coordinate.

Quadratic functions of inverse charge describe the charge dependence of the measured arrival time for four silicon-photomultiplier channels on two reference bars. These functions provide an empirical time-walk calibration over the measured charge range. Clock-corrected profiles provide the reference for the companion dual-threshold study; archived fits are retained to document the change in calibration procedure. Timing-difference distributions illustrate the validation strategy. Their interpretation separates intrinsic reference-detector resolution from residual synchronization effects. This draft presents the results and the selection, uncertainty and reproducibility checks that remain before publication.

Beam-test setup and the reference signal



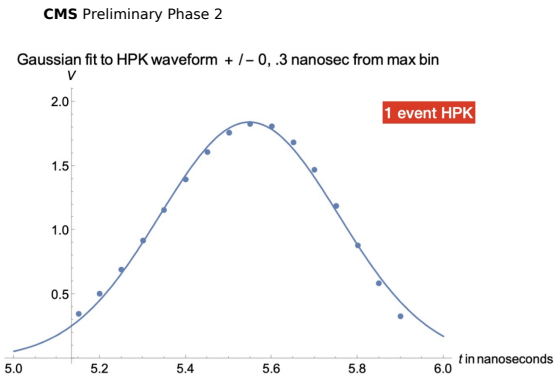
The September 2025 beam test used scintillator bars read out at both ends by silicon photomultipliers (SiPMs). A downstream MCP-PMT provides a fast reference for a restricted region of the bars.

The BTL front-end readout and oscilloscope use independent time coordinates. The scope trigger reaches TOFHIR channel 192; a separate 160 MHz clock link can also be recorded by the scope.

The reference-bar calibration uses the highlighted lower readout path.

Original setup illustration. Detector and readout context: Ref. [4]; the run-specific setup is described in this study.

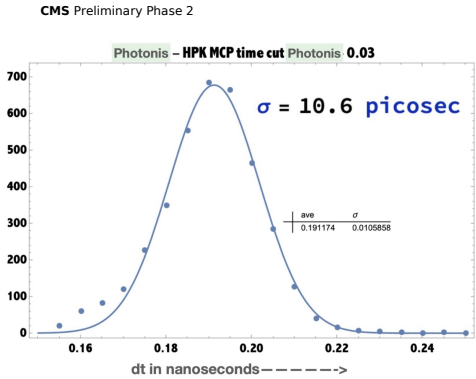
Reconstructing the MCP pulse time



The figure shows a Hamamatsu MCP waveform and its fitted pulse template. The fit estimates the pulse centroid rather than using the threshold-crossing time alone, reducing sensitivity to pulse amplitude.

The scope-trigger time identifies the event; the fitted peak supplies its relative reference time. Their difference is measured within the same oscilloscope waveform.

Intrinsic reference timing is a separate measurement

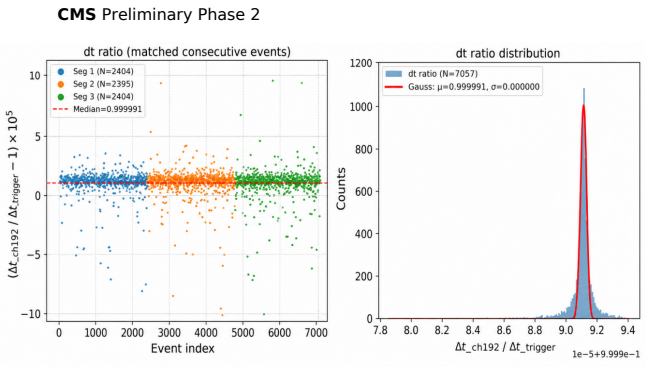


The distribution shows the difference between the fitted times of two MCP-PMTs recorded by the oscilloscope. Its measured width is 10.6 ps.

Assuming independent errors and a 7.5 ps PHOTONIS contribution gives $\sigma_{\mathrm{HPK}} \simeq \sqrt{10.6^2 - 7.5^2} = 7.5$ ps.

This scope-only estimate does not include synchronization to the BTL readout.

Matching the same particles between readouts



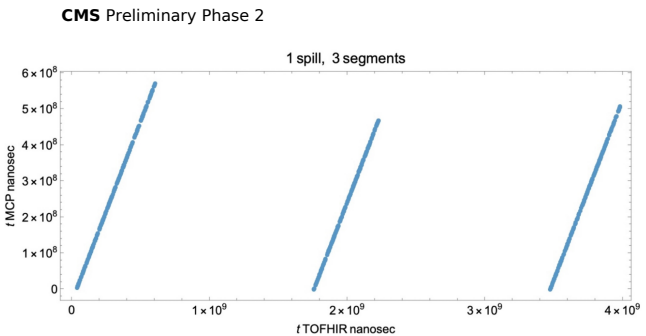
The matching procedure compares consecutive trigger intervals measured by channel 192 with those recorded by the scope. Correlated intervals identify candidate matches despite the independent clocks.

The interval-ratio distribution tests matching consistency. It is not a measurement of the picosecond resolution of the reconstructed particle time.

Only uniquely matched events enter the reference calibration.

GitHub: trigger-matching implementation. `dedup_triggers`: duplicate removal. `match_cumulative_time`: matching and tolerance.

Clock-rate alignment within acquisition segments

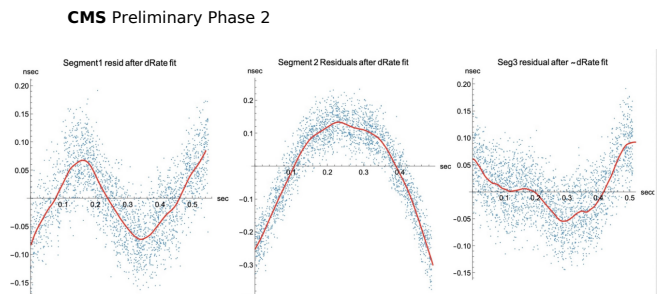


The figure compares matched scope-trigger times with TOFHIR trigger times. A nearly linear relation describes the overall clock-rate difference within an acquisition segment.

Both an offset and a slope are needed. Even a slope close to one matters over a long time interval. Clock resets require separate segment coordinates.

Archived clock-relation plot. It illustrates synchronization, not charge-dependent time walk.

Correcting smooth residual clock drift



After the linear clock-rate correction, the time residual varies within each of three segments. Points represent matched events; red curves show the smooth trend estimated by LOWESS, a locally weighted regression [8]. Adding the fitted trend to the linear prediction reconstructs the expected channel-192 trigger time. The reference difference subtracts this expected time once from the SiPM time.

Archived residuals in ns versus segment time in seconds. These are pre-LOWESS residuals, not final timing resolutions. GitHub: clock prediction; reference subtraction.

A single, explicit time convention

Let $g(u)$ map a scope-trigger timestamp u into the TOFHIR time coordinate, including the linear clock relation and the fitted smooth drift. Define the within-waveform difference

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}.$$

The reference difference implemented in the calibration is

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - \delta_{\mathrm{MCP}},$$

up to a fixed cable and geometry offset. The clock-rate correction for the short within-waveform interval is negligible and is omitted.

The charge-dependent component of Δt_i is the time walk. An arbitrary constant does not change a shape calibration.

GitHub implementation: per-file subtraction; combined-run subtraction. Both per-file and combined-run calculations subtract (`mcp_peak - mcp_trig`).

Reference bars and calibration selection

Upstream bar	SiPM channels	Role
7	133, 154	First reference pair
8	137, 150	Second reference pair

- Select coincident SiPM hits and a unique MCP-associated event, using channel 192 as the TOFHIR indicator.
- Require a successfully reconstructed MCP pulse and consistent peak-to-trigger timing. This is a waveform-quality cut, not a particle transit-time measurement inside the MCP.
- Use Q, the TOFHIR ADC charge proxy. It is not a deposited energy in MeV.
- Fit the profile of Δt_i versus $1/Q$ over the retained charge range.

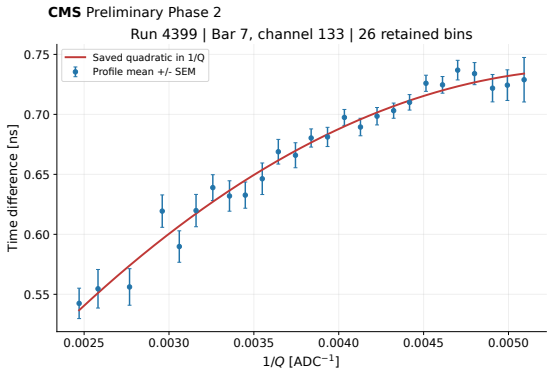
GitHub implementation:

Coincidence and walk-sample selection.

MCP mapping and optional cuts.

Peak-to-trigger timing selector.

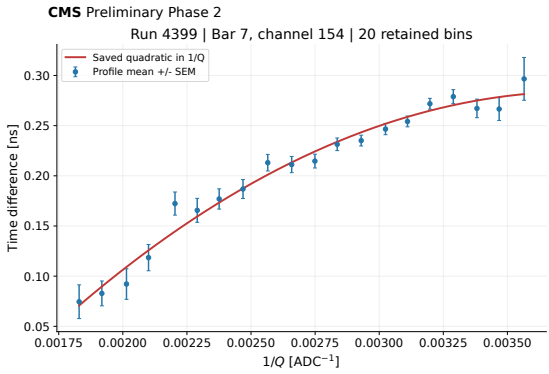
Archived reference: bar 7, channel 133



Run 4399 events are grouped in inverse-charge bins. Points show mean time differences with standard errors of the mean (SEM). The curve is the archived fit, $W(Q) = a + b/Q + c/Q^2$. There are 26 retained profile points. Times include the historical display offset. These archived means use measured channel-192 timing. An unweighted fit to the means reproduces the archived coefficients; SEM bars are not fit weights.

Archived profile and coefficients, displayed without refitting. The display convention includes a +26 ns shift.

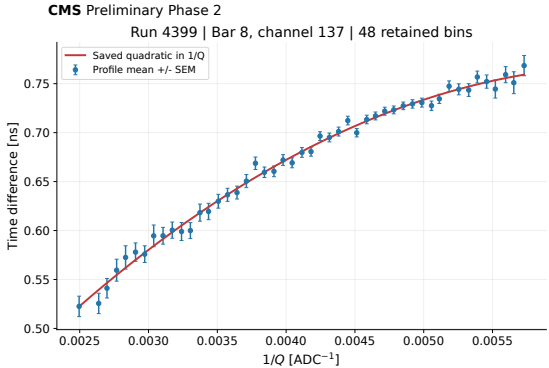
Archived reference: bar 7, channel 154



The opposite end of bar 7 is calibrated independently. Points show the mean time difference and SEM in 20 retained inverse-charge bins.
The red curve is the archived quadratic in $1/Q$, not a quadratic in Q. Channel-dependent coefficients allow for different pulse responses at the two bar ends. The archived fit is unweighted and uses measured channel-192 timing. The retained bin counts are channel dependent.

Run 4399 archived calibration, redrawn without refitting. Same +26 ns display convention as channel 133.

Archived reference: bar 8, channel 137

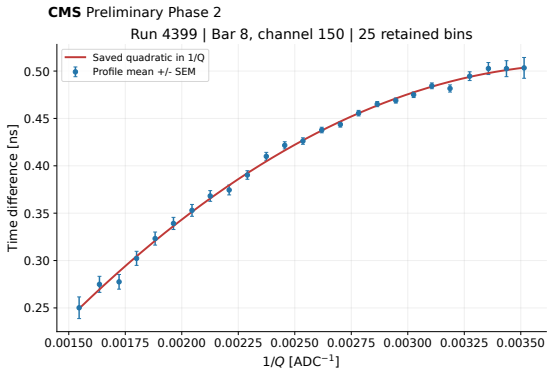


The channel-137 time difference increases as inverse charge increases: smaller signals cross a fixed threshold later.

Points are profile means with SEM; the red line is the archived quadratic fit to the profile. There are 48 retained points. The correction applies within the measured charge range; extrapolation is not validated. This unweighted result uses measured channel-192 timing and is retained for comparison with the clock-corrected reference shown next.

Run 4399 archived calibration. Coefficients and profile points are included with the draft.

Archived reference: bar 8, channel 150



The opposite end of bar 8 has its own charge-dependent delay. The 25 points show the mean timing difference and SEM; the curve is the archived quadratic fit to the profile.
The four curves demonstrate the calibration procedure. Each channel retains its original number of profile points and its original binning. This is the archived, unweighted fit using measured channel-192 timing.

Run 4399 archived calibration. Coefficient units are ns, ns ADC and ns ADC^2 for a, b and c.

Clock-corrected reference shared with Note 2

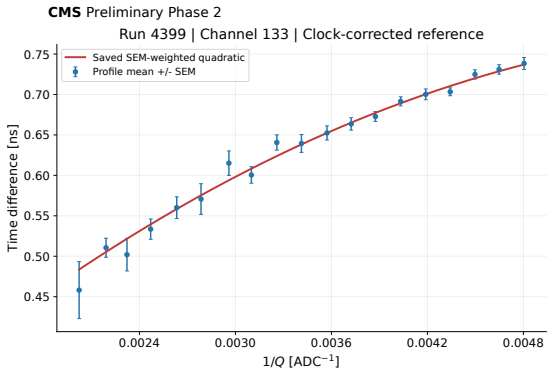
The following results use the expected trigger time $g(u)$ reconstructed from the segment-wise linear clock relation and LOWESS drift:

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - (t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}).$$

- Run 4399: exactly five TOFHIR hits, comprising the target upstream pair, downstream channels 234 and 243, and channel 192, with a unique MCP match.
- Require finite, positive charge and slope-related quality quantities; retain Q at or above the left half-maximum boundary of the spectrum peak.
- Form 20 uniform $1/Q$ bins; retain times within $5(1.4826\,\mathrm{MAD})$ of each bin's median, with at least five entries. Bars show the SEM after clipping.
- Fit $W(Q) = a + b/Q + c/Q^2$ with an objective weighted by $1/\mathrm{SEM}^2$. The next four figures use the same reference coefficients as Note 2.

The displayed $+26$ ns constant changes neither the shape nor the saved coefficients.

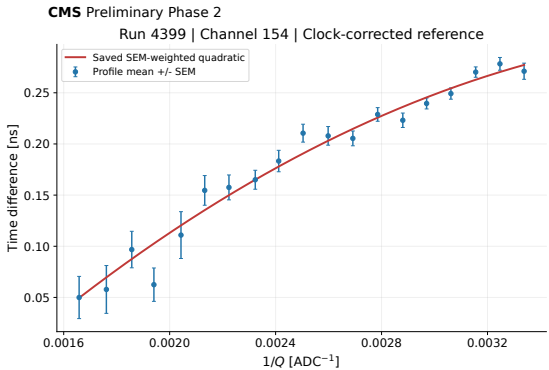
Clock-corrected reference: bar 7, channel 133



The time difference now uses the reconstructed trigger time rather than the measured channel-192 timestamp, suppressing the fitted clock drift.
Points show mean time differences with SEM in 19 retained inverse-charge bins.
The red curve is the SEM-weighted quadratic used in Note 2.
The curve is restricted to retained charge bins and the range between the outer profile means. Gaps from missing bins are left blank.

Run 4399; the reference used in Note 2, redrawn without refitting. A constant +26 ns shift is applied only for display.

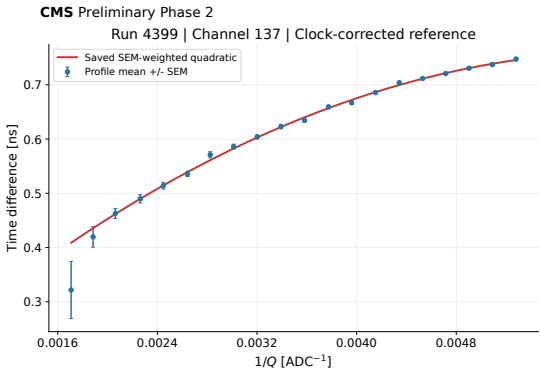
Clock-corrected reference: bar 7, channel 154



The same clock reconstruction and profiling procedure calibrates the opposite end of bar 7. Independent coefficients describe its charge-dependent timing response. The 19 points are retained-bin means; vertical bars show SEM after timing clipping. The red curve is the SEM-weighted quadratic in $1/Q$.

The reference used in Note 2. Coefficients and profile points accompany this note; the $+26$ ns shift is applied only for display.

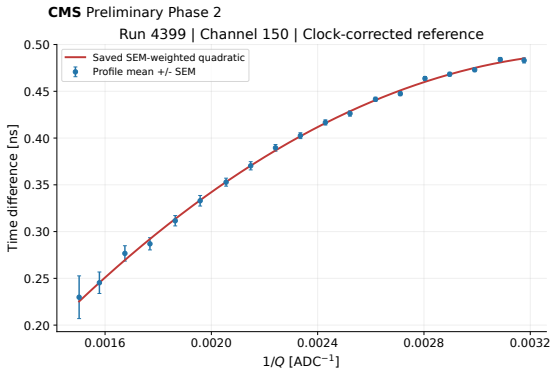
Clock-corrected reference: bar 8, channel 137



There are 20 retained inverse-charge bins for this channel. Smaller charge gives a larger timing delay, as expected for fixed-threshold readout of pulses with different amplitudes.
Points and bars are the mean and SEM after clipping. The saved quadratic is the MCP benchmark for the channel-137 inverse-slope comparison in Note 2.

Run 4399; expected-trigger timing; refitted after clock-quality selection. The +26 ns shift affects only the vertical display origin.

Clock-corrected reference: bar 8, channel 150



The channel-150 reference uses 19 retained profile bins and the same clock-corrected time definition as the other three channels.
Bars represent SEM, not the spread of individual event times. The red line is the SEM-weighted quadratic in inverse charge. These four results connect the MCP calibration in this note to the AWC shape comparison in Note 2.

Run 4399; coefficients are preserved in their original ns convention. Only the figure receives the constant +26 ns display shift.

Archived and clock-corrected shapes are close

The comparison uses both fitted functions in charge intervals populated in both samples, after removing a constant offset. No multiplicative scaling is fitted.

Channel	Shape RMS difference [ps]	Maximum absolute difference [ps]
133	3.20	10.63
154	4.29	9.68
137	1.33	3.22
150	4.91	10.47

The current reference identifies clock resets from finite timestamps and applies a clock-only quality requirement. The archived and current versions also differ in trigger reconstruction, selection, clipping and fit weights.

These are differences between calibration functions, not timing resolutions or a statistical-equivalence test. Note 2 uses the clock-corrected version.

Clock quality before timing validation

Segment boundaries are identified from consecutive finite scope timestamps, including resets across missing entries. Nonfinite trigger times have no valid mapping.

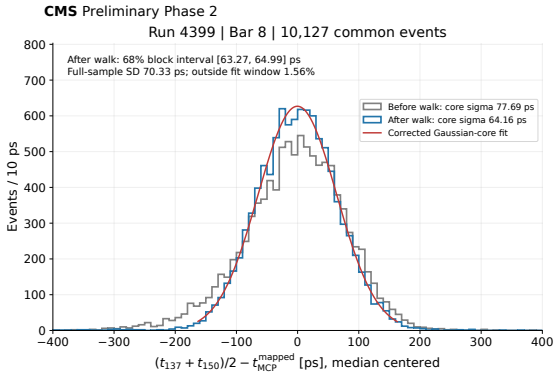
For each topology, file and segment, fit channel 192 against the scope trigger. Reject the whole group if the line-residual scale $1.4826\,\mathrm{MAD}$ exceeds 1 ns, before LOWESS. This rule uses clocks only, not SiPM arrival times.

Six bar-8 groups with finite trigger times fail this cut. Changing the threshold from 1 ns to 1 μs retains the same groups. Suspected one-event association offsets are excluded; MCP rows are not shifted to repair them.

Refit the channel walk curves after this selection. For the following timing comparison, both ends must pass the quality and lower-charge cuts. Each end must also lie within its populated reference bins and outer profile means. No additional selection is made on either timing-difference observable.

GitHub: clock-quality selection.

Timing validation against the MCP reference



For 10127 events, distributions show bar-average minus mapped MCP time before and after walk correction, with fixed clock mapping.
The core width changes from 77.69 to 64.16 ps. The corrected full-sample SD is 70.33 ps; 1.56% of events lie outside the fit window.
The conditional 16–84% interval is [63.27, 64.99] ps (400 file/segment-block bootstraps, fixed calibrations). Detector, MCP and residual clock effects contribute.

Truncated-normal fit: median $\pm 2.5(1.4826\,\mathrm{MAD})$. Median-centered display; full-sample SD; no segment-wise recentering.

What these timing widths establish

The reproduced distributions compare timing performance before and after correction under fixed selections and calibrations. They do not measure intrinsic bar resolution or isolate synchronization jitter. At fixed position, for end timing errors e_L, e_R,

$$\mathrm{Var}\Big(\frac{e_L + e_R}{2}\Big) = \frac{\mathrm{Var}(e_L - e_R)}{4} + \mathrm{Cov}(e_L, e_R).$$

Halving an end-difference width requires controlled position effects and negligible end-error covariance. Shared MCP and clock errors also prevent inferring detector covariance directly from the measured reference residuals.

Separately fitted core widths cannot be treated as variances of a common model to infer synchronization jitter by quadrature subtraction. This version reports same-event timing validation with conditional statistical intervals. Intrinsic bar resolution and synchronization-jitter extraction are outside its scope. Position, covariance, segment-offset and calibration systematics are not included in the quoted intervals.

Reference-calibration result and scope

- A trigger link enables calibration with common events across independent BTL and oscilloscope time coordinates.
- A linear clock relation plus a smooth residual correction defines the reference-time mapping.
- Four clock-corrected inverse-charge profiles and quadratic functions provide the reference used in Note 2. The archived fits remain as a comparison; profile points, SEM and both coefficient sets accompany this draft.
- Intrinsic MCP timing, reference-bar timing and cross-system synchronization are different performance quantities.
- Charge range and operating conditions limit transfer to other samples; a constant time offset is not a walk-shape difference.

For the same selected events, walk correction reduces the bar–MCP core width from 77.69 to 64.16 ps. This is a conditional timing-validation result, not an intrinsic-resolution measurement or a complete uncertainty budget.

References (1/2)

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

References (2/2)

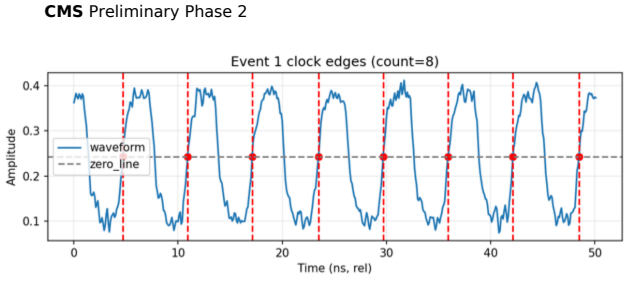
[5] S. White, Nov. 2025 BTL TB Analysis mtg presentation, https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf

[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

Backup: clock-phase reconstruction



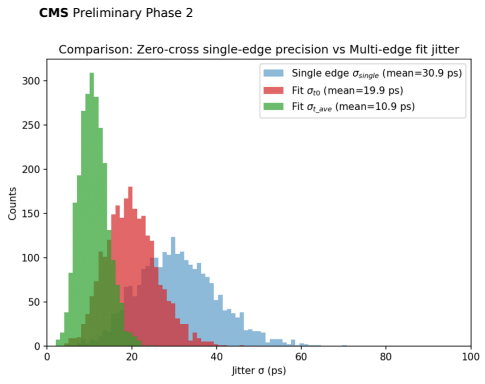
The reference clock is sampled by the oscilloscope. Several rising-edge crossings can be reconstructed within a waveform.

Fitting edge time against edge index estimates the clock phase from multiple cycles. Evaluating the fit near the mean edge index avoids unnecessary extrapolation.

This is an alternative synchronization diagnostic, distinct from the trigger-link method used for the archived reference curves.

Archived illustration. Correlated edge errors preclude assuming a $1/\sqrt{N}$ improvement in precision.

Backup: comparing clock estimators



The recorded MCP time is compared with three clock-phase estimators: a single edge, a regression intercept and the fitted time at the mean edge index. The distributions illustrate the dependence on the estimator. Their widths include all contributions present in this measurement and do not directly measure absolute clock accuracy. The underlying data, common selections and width definitions have not been established. This internal discussion figure is excluded from the publication candidate.

Archived clock-estimator diagnostic, retained for discussion.